

Flight Control Circuit Organization in Drosophila BANC Connectome

Major Findings

A. Circuit Architecture

5 Functional Modules Discovered

2,080 Premotor Interneurons Organized into Power and Steering Networks

POWER MODULE

Cluster 4

370 interneurons

STEERING MODULES

Clusters 1, 2, 3, 5

1,710 interneurons

Individual Clusters:

- C1: 335 INs → b2, b1, iii4
- C2: 539 INs → iii4, hg4, hg1
- C3: 460 INs → i2, i1, b3
- C4: 370 INs → DLM1, DLM5, DVM1A
- C5: 376 INs → ps1, iii1, iii3

Power: Strong connections | Steering: Distributed control

B. Network Modularity

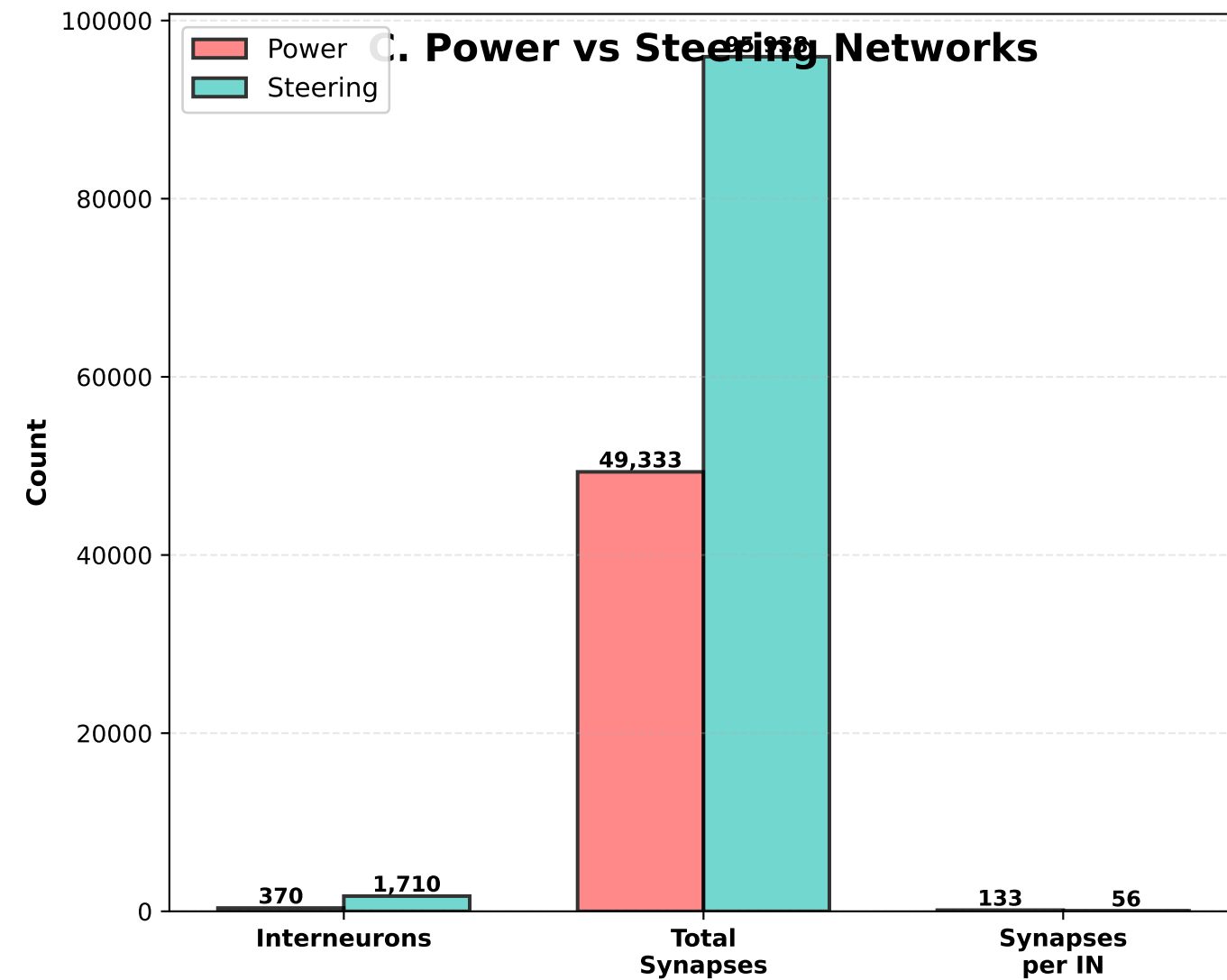
0.354

Modularity Score

MODERATE Functional Separation

Cluster Connectivity Strength:

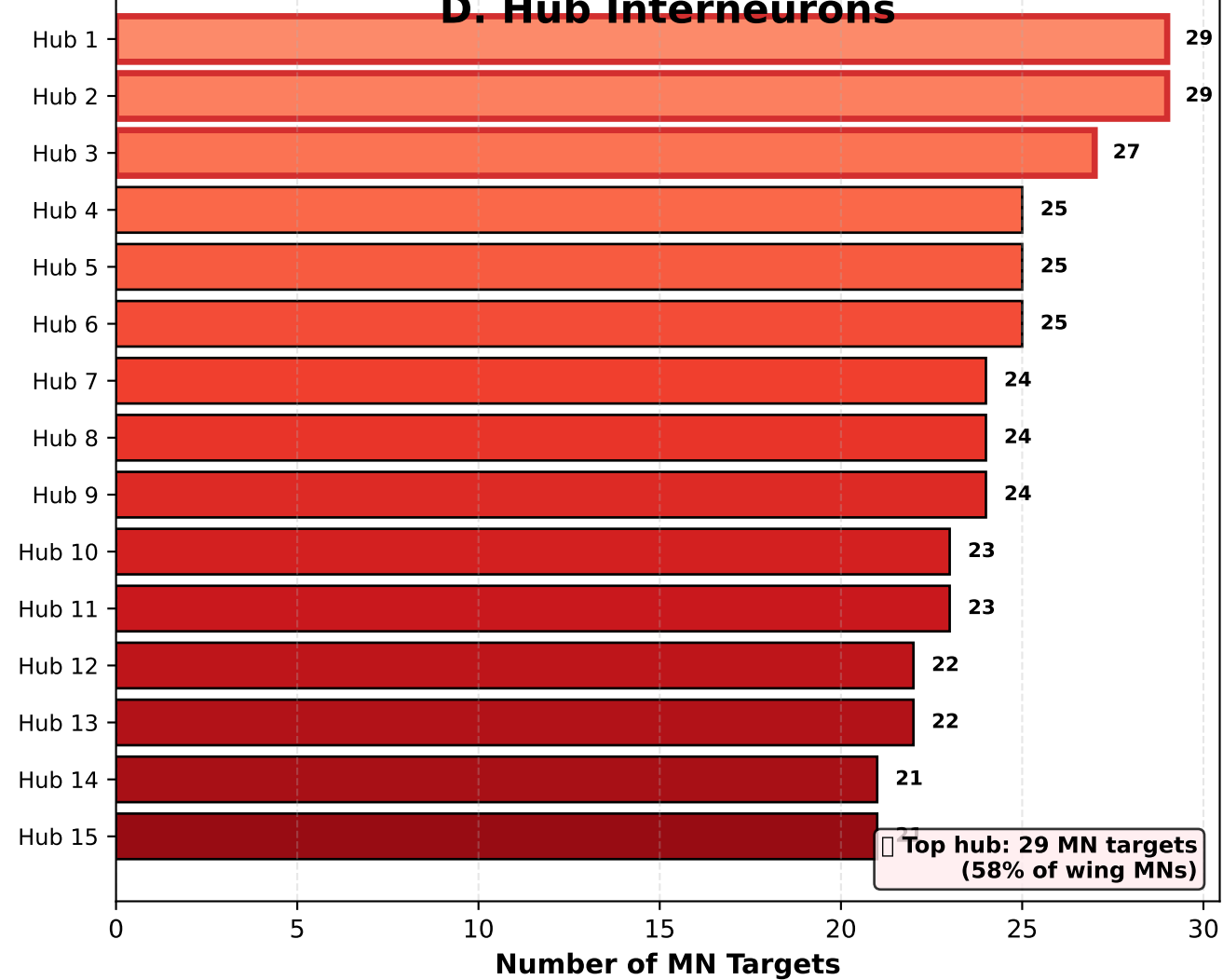
C. Power vs Steering Networks



Steering has 1.9× more total synapses

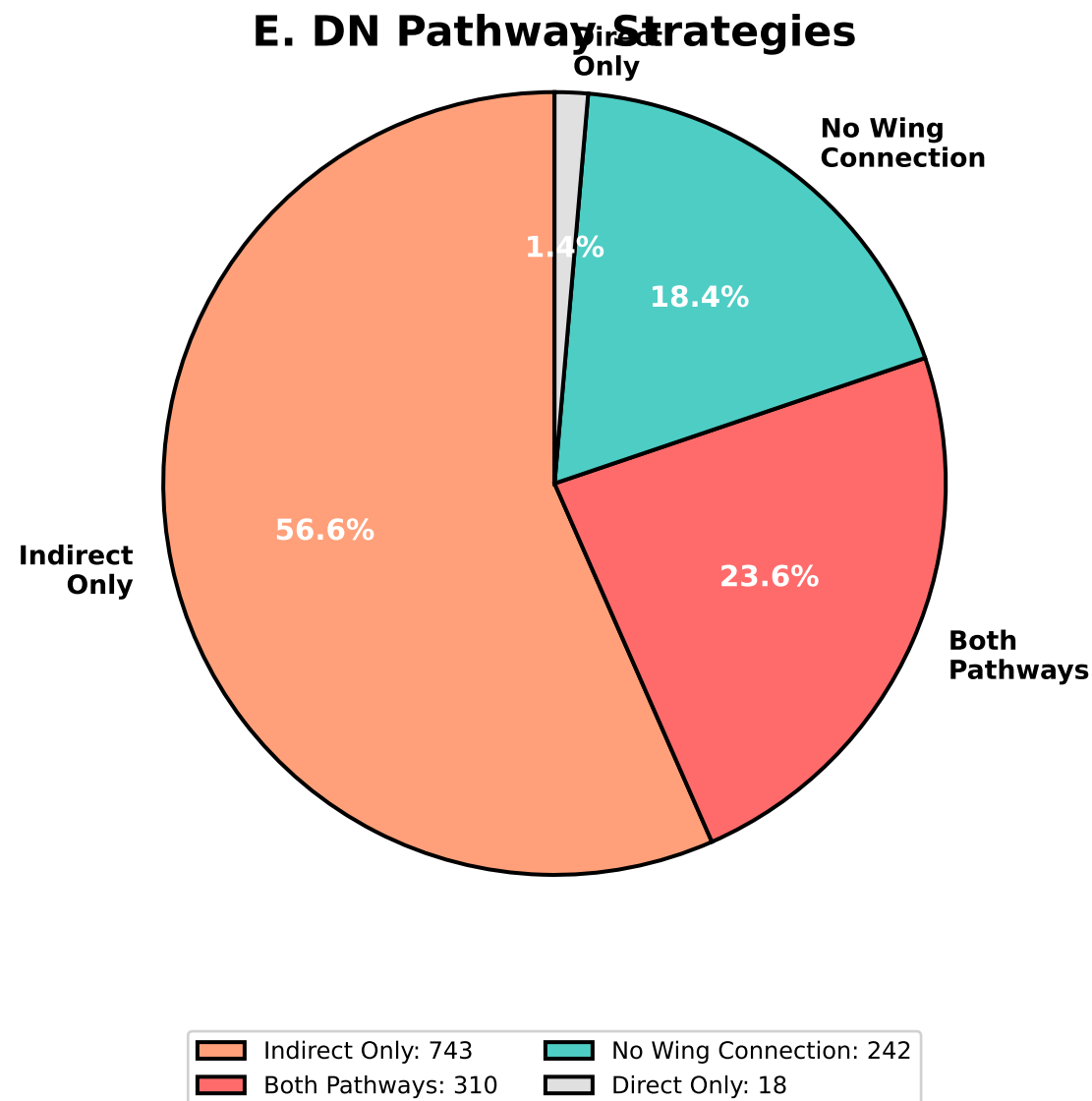
Top 15 of 2,080 Hub Neurons

D. Hub Interneurons



Top hub: 29 MN targets
(58% of wing MNs)

E. DN Pathway Strategies



F. Key Statistics

CIRCUIT SUMMARY

Premotor Network:

- Total INs: 2,080
- Connections: 14,077
- Synapses: 145,271

Functional Modules:

- Total: 5
- Power: 1 cluster (370 INs)
- Steering: 4 clusters (1,710 INs)

Hub Neurons:

- Total identified: 2,080
- High centrality: 829
- Max targets: 29

Integration:

- Integrative INs: 335
- Percentage: 16.1%

Descending Neurons:

- Analyzed: 1313
- Both pathways: 310
- Direct only: 18
- Indirect only: 743